Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: 20260206_171419_resnet50_classifier.pt

Attack Type: FGSM

Images Evaluated: 10

Attack Success Rate	Original Confidence	Adversarial Confidence	Execution Time
90.0%	33.76%	46.76%	5.31s Accuracy drop: -13.01%

Threat Level Assessment

HIGH RISK

The FGSM attack altered model predictions on 90.0% of the evaluated images. Average confidence shifted from 33.76% to 46.76%, indicating the current robustness posture is high risk.

Assessment Narrative

Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 90.0%. Average original accuracy: 33.76%, Average adversarial accuracy: 46.76%. The attack successfully fooled the model in 9 out of 10 cases.

Detailed Analysis

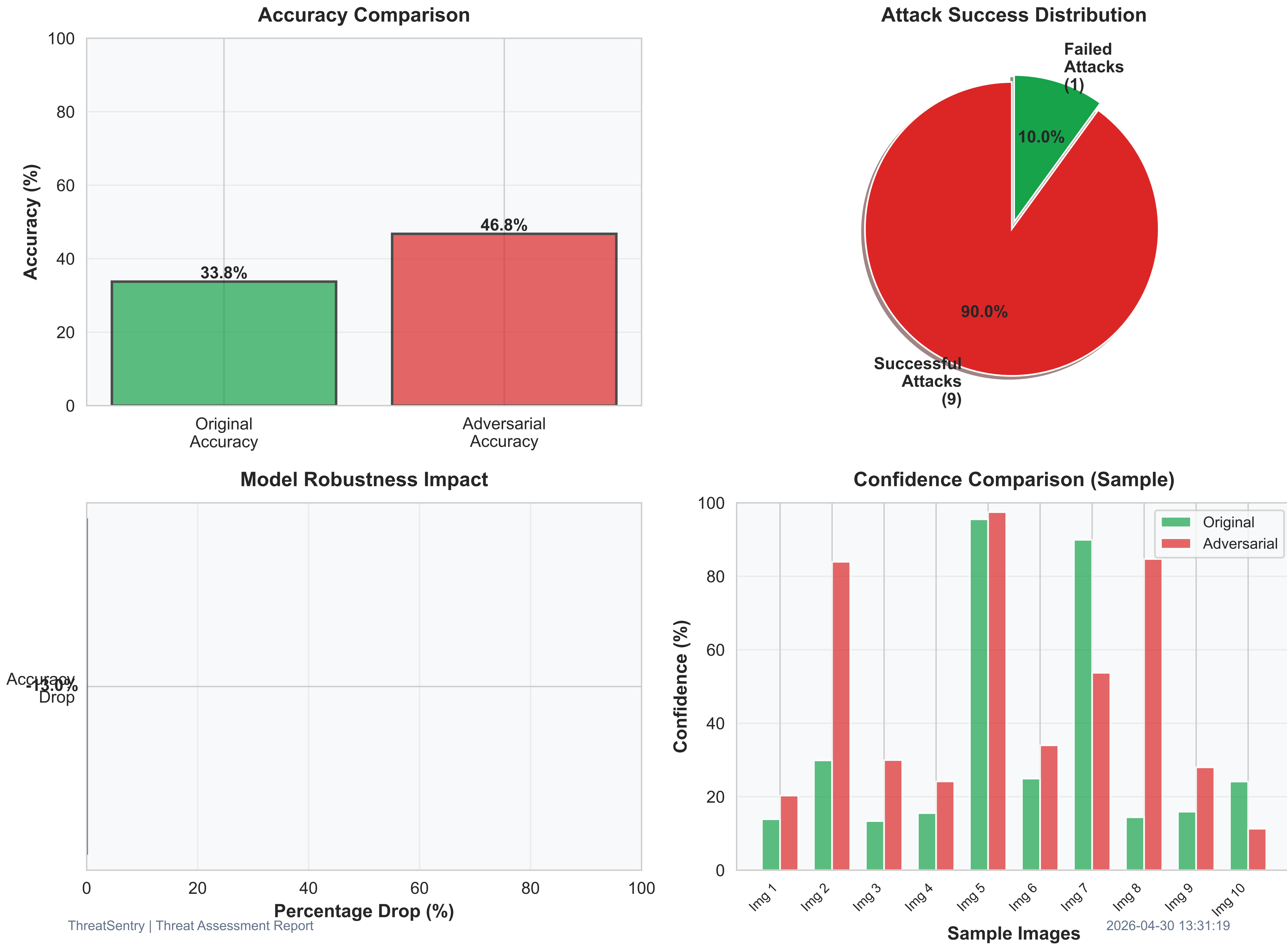
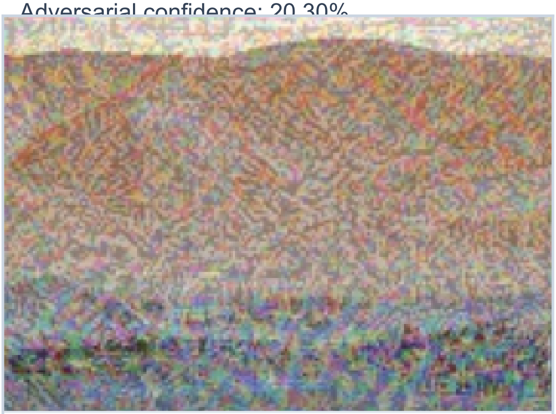


Image Evidence and Predictions

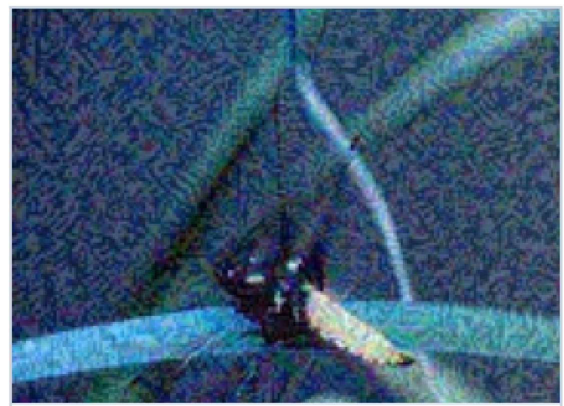
Images 1-4 of 10

Attack Successful
Original image: rock_crab.JPEG
Adversarial image: rock_crab.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: rock crab, Cancer irroratus
Original confidence: 13.88%
After attack predicted as: jigsaw puzzle
Adversarial confidence: 20.30%

Attack Successful
Original image: breastplate.JPEG
Adversarial image: breastplate.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: mortar
Original confidence: 24.94%
After attack predicted as: jigsaw puzzle
Adversarial confidence: 34.01%

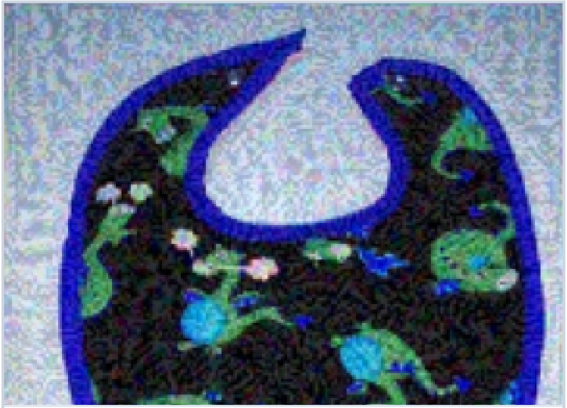


Attack Successful
Original image: valley.JPEG
Adversarial image: valley.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: stole
Original confidence: 24.14%
After attack predicted as: brass, memorial tablet, plaque
Adversarial confidence: 11.29%

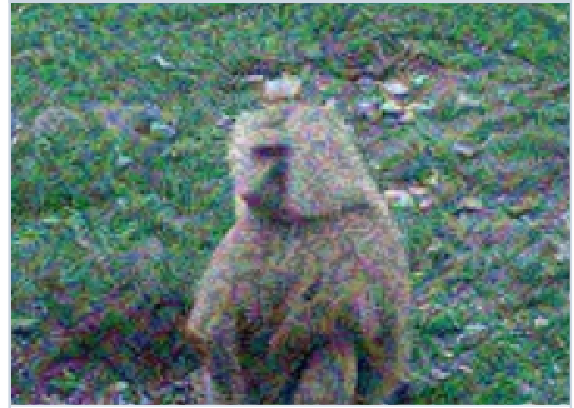


Attack Successful
Original image: dragonfly.JPEG
Adversarial image: dragonfly.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: tarantula
Original confidence: 15.54%
After attack predicted as: starfish, sea star
Adversarial confidence: 24.17%

Image Evidence and Predictions



Attack Blocked
Original image: bib.JPEG
Adversarial image: bib.JPEG (attacked)
Outcome: Attack Blocked
Model predicted as: bib
Original confidence: 95.52%
After attack predicted as: bib
Adversarial confidence: 97.48%



Attack Successful
Original image: baboon.JPEG
Adversarial image: baboon.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: baboon
Original confidence: 89.94%
After attack predicted as: American alligator, Alligator mississipiensis
Adversarial confidence: 53.74%

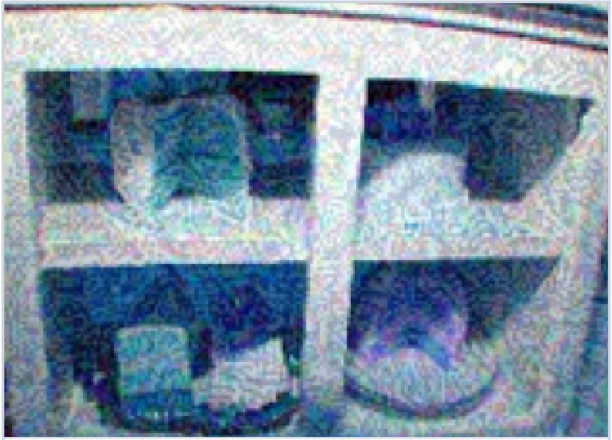
Attack Successful
Original image: church.JPEG
Adversarial image: church.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: chainlink fence
Original confidence: 29.89%
After attack predicted as: jigsaw puzzle
Adversarial confidence: 83.96%

Attack Successful
Original image: pot.JPEG
Adversarial image: pot.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: poncho
Original confidence: 14.41%
After attack predicted as: spider web, spider's web
Adversarial confidence: 84.72%

Image Evidence and Predictions

Images 9-10 of 10

Attack Successful
Original image: knee_pad.JPEG
Adversarial image: knee_pad.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: fountain
Original confidence: 15.93%
After attack predicted as: jigsaw puzzle
Adversarial confidence: 28.00%



Attack Successful
Original image: bakery.JPEG
Adversarial image: bakery.JPEG (attacked)
Outcome: Attack Successful
Model predicted as: pillow
Original confidence: 13.38%
After attack predicted as: envelope
Adversarial confidence: 29.98%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 90.0%. Average original accuracy: 33.76%, Average adversarial accuracy: 46.76%. The attack successfully fooled the model in 9 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques